

Supplementary Information

CrossPiezo: Elite-Tail Instability in Cross-Database Piezoelectric Screening

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1 Supplementary notes

1.1 Portfolio estimand definition (Note 1)

The portfolio benchmark reports two related but distinct quantities. *Worst-source recall* is a material-level (micro) metric: it counts how many of the top- q^*n materials selected by a strategy are also present in the JARVIS and MP elite sets, divided by q^*n , with no group weighting. *Paired difference in recall* is computed by first evaluating a grouped worst-source recall within each reduced-formula group, then taking the mean difference across groups with a group-level paired bootstrap confidence interval. Because each reduced-formula group contains only a handful of materials, the group-mean differences are small even when the material-level recall improves substantially. Consequently, the ΔRecall column in Table S3 should not be interpreted as the arithmetic difference of the Recall column; the two columns answer different statistical questions. The portfolio analysis is therefore exploratory and is not presented in the main text as a validated method comparison.

1.2 Structure matching and manual audit (Note 2)

The CrossPiezo-Invariant-v1 panel was constructed from relaxed CIFs shared by the Materials Project (MP) and JARVIS. The matching pipeline is: (i) load all processed MP and JARVIS piezoelectric records; (ii) compute structure-level overlaps by compositional reduced formula and relaxed-lattice similarity; (iii) apply a frozen similarity threshold to obtain candidate pairs; (iv) perform a response-blind manual audit of the top-matched candidates using a frozen rule set. The audit was response-blind: the auditor did not see piezoelectric response values when deciding whether a candidate pair represented the same or a distinct structure. All audit decisions, exclusion reasons and ambiguous cases are recorded in the frozen panel-membership file. P0 ($n = 573$) is the full set of accepted structure matches; P2 ($n = 207$) is the subset passing a tighter similarity threshold.

1.3 Tensor invariants and conventions (Note 3)

All piezoelectric response values are piezoelectric stress tensors $\mathbf{e}$ in C/m² with full Cartesian $3 \times 3 \times 3$ components. The three coordinate-invariant scalars are:

- F1, Cartesian Frobenius norm $\|\mathbf{e}\|_F$;
- F3, true collinear longitudinal maximum $\max_{\|\mathbf{n}\|=1} |n_i e_{ijk} n_j n_k|$ computed by numerical optimisation over the unit sphere;
- F4, Kelvin/Mandel operator norm on symmetric strain, computed from the Mandel representation of the 6×3 piezoelectric tensor.

Voigt-to-Cartesian conversion follows the standard convention $11 \rightarrow xx$, $22 \rightarrow yy$, $33 \rightarrow zz$, $23 \rightarrow yz$, $13 \rightarrow xz$, $12 \rightarrow xy$ with factor-of-two handling for shear components as appropriate for the stress tensor. No sign conventions, unit conversions or coordinate rotations were applied silently; all transformations are recorded in the repository transformation history.

1.4 Bootstrap, quantile and tie handling (Note 4)

Screening-resolution curves are evaluated at quantiles $q = 1\%, 2\%, \dots, 50\%$. The chance-adjusted Jaccard overlap is $\tilde{J}_q = (J_q - \mathbb{E}[J_q]) / (1 - \mathbb{E}[J_q])$, where $\mathbb{E}[J_q]$ is the exact hypergeometric expectation under independent rankings. Simultaneous 95% confidence bands use a paired bootstrap with a

studentized sup-norm critical value. Bootstrap resampling is by reduced-formula group to respect the dependence among entries sharing a composition; the random seed, number of replicates and studentization details are recorded in the Phase 7C configuration. Ties in source rankings are broken by stable sorting; duplicate reduced formulae are retained in the ranking but noted in the panel metadata. The persistent-onset criterion requires the lower confidence bound to exceed a threshold δ for at least five consecutive quantiles.

2 Supplementary figures

3 Supplementary tables

Table 1: Corrected high-response sensitivity for P0 F1 at $f = 0.10$ and $f = 0.25$.

Method & f & N & au & CI & nAUCC & CI						
Dual-high	10%	58	0.035	[-0.193, 0.239]	-0.046	[-0.118, 0.044]
JARVIS anchor	10%	57	0.209	[-0.062, 0.374]	0.053	[-0.061, 0.191]
MP anchor	10%	57	-0.053	[-0.235, 0.137]	-0.018	[-0.104, 0.090]
Pooled (exploratory)	10%	58	—	[-, -]	-0.155	[-, -]
Dual-high	25%	143	0.070	[-0.046, 0.181]	0.031	[-0.034, 0.102]
JARVIS anchor	25%	143	-0.019	[-0.141, 0.105]	-0.006	[-0.070, 0.070]
MP anchor	25%	143	0.039	[-0.083, 0.154]	0.007	[-0.052, 0.078]
Pooled (exploratory)	25%	144	—	[-, -]	-0.100	[-, -]

Table 2: Property-control comparison on P0. The same-field copy flag is raised when more than 5% of finite matched pairs have identical JARVIS/MP values.

Attribute & N & au & 95% CI & nAUCC & 95% CI & $\Delta\tau$ & Δ nAUCC & Copy flag								
Volume	573	0.967	[0.963, 0.971]	0.918	[0.901, 0.936]	0.722	0.813	&
Band gap	573	0.908	[0.890, 0.923]	0.897	[0.864, 0.908]	0.663	0.792	&
Energy above hull	573	0.611	[0.547, 0.670]	0.533	[0.455, 0.592]	0.366	0.428	& ✓
Dielectric trace	573	0.482	[0.418, 0.547]	0.286	[0.228, 0.344]	0.236	0.181	&

Table 3: Equal-budget ($b = 1.0$) portfolio results on P0 F1 ($q^* = 10\%$). Paired differences are versus the better single-source baseline and are computed as group-mean differences, not as arithmetic differences of the material-level Recall column (see Note 1).

Strategy & Recall & NDCG & Coverage & Δ Recall & 95% CI						
JARVIS only	0.140	0.646	0.099	0.000	[0.000, 0.000]	
MP only	0.140	0.673	0.099	0.014	[-0.022, 0.050]	
Average percentile	0.456	0.915	0.099	0.014	[-0.012, 0.042]	
Borda	0.456	0.915	0.099	0.014	[-0.012, 0.042]	
Maximin	0.386	0.909	0.099	0.014	[-0.016, 0.044]	
Intersection first	0.456	0.915	0.099	0.014	[-0.012, 0.042]	
Balanced union	0.526	0.902	0.099	0.012	[-0.012, 0.036]	
Disagreement abstention	0.456	0.915	0.099	0.014	[-0.012, 0.042]	
Minimax oracle	0.474	0.915	0.099	0.014	[-0.012, 0.040]	

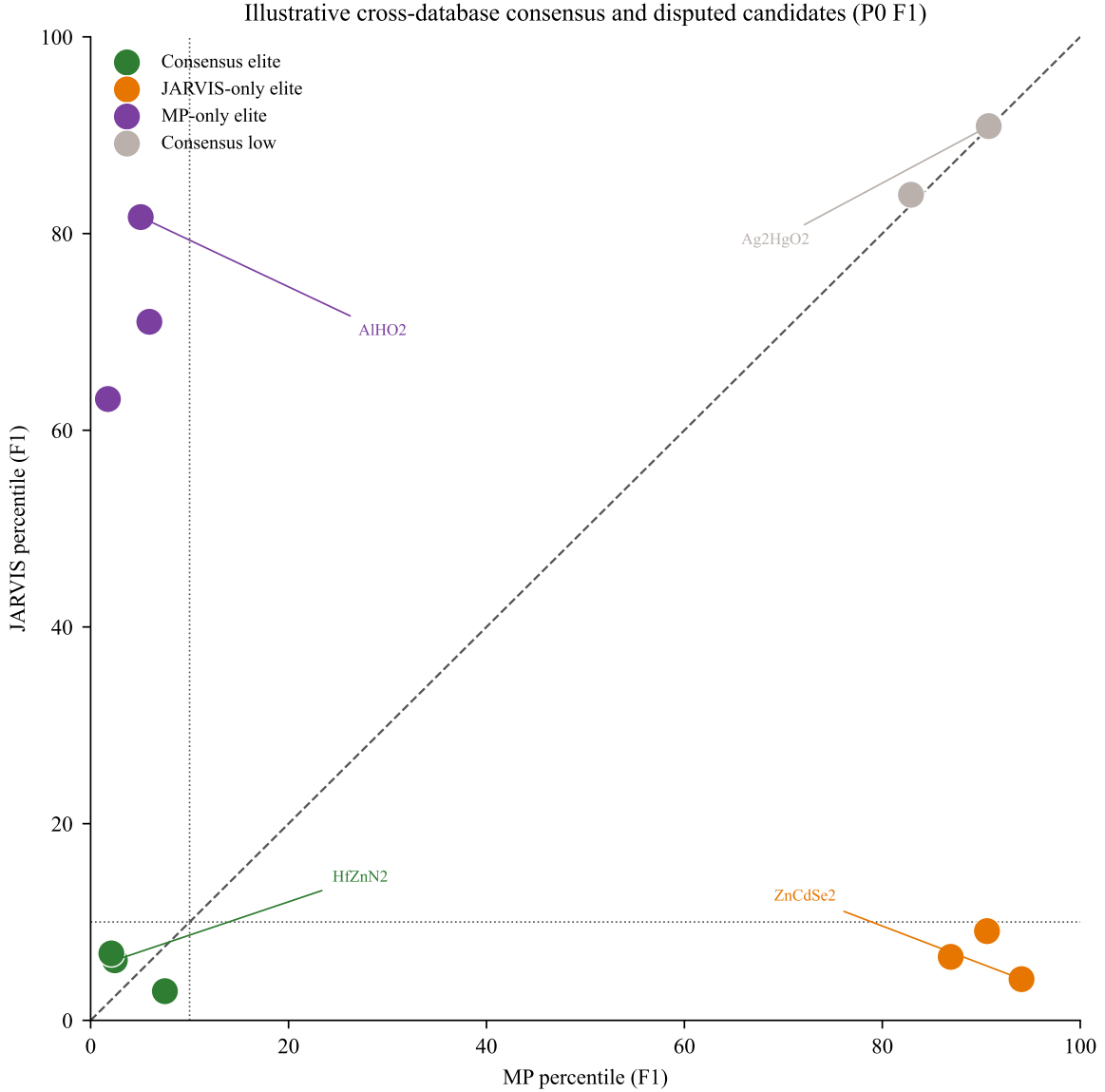


Figure 1: Illustrative cross-database consensus and disputed candidates on P0 F1. Points show JARVIS versus MP percentiles for consensus elite (green), JARVIS-only elite (orange), MP-only elite (purple) and consensus low (grey) materials. The dashed diagonal marks equal ranking; dotted lines mark the 10% elite threshold. These examples are illustrative and are not adjudicated as physically correct or incorrect.

Table 4: Portfolio worst-source recall on P0 F1 across budget factors $b = 1.0, 1.5, 2.0$.

Strategy & $b = 1.0$ & $b = 1.5$ & $b = 2.0$
JARVIS only & 0.140 & 0.281 & 0.333
MP only & 0.140 & 0.228 & 0.316
Average percentile & 0.456 & 0.526 & 0.579
Maximin & 0.386 & 0.456 & 0.491
Balanced union & 0.526 & 0.789 & 1.000

Portfolio cost-coverage frontier on P0 F1 ($q^* = 10\%$)

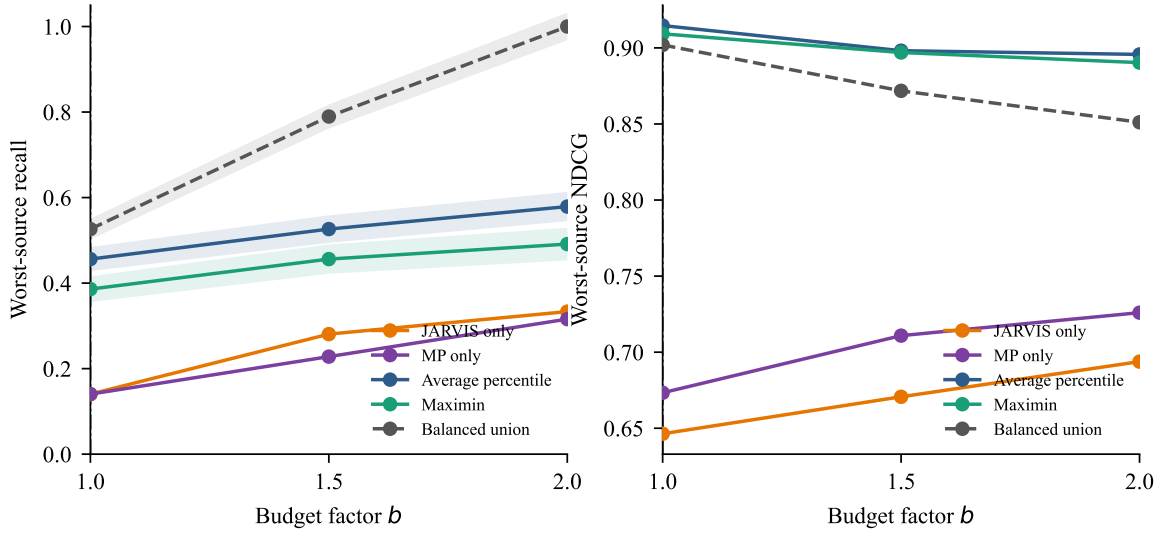


Figure 2: Portfolio cost-coverage frontier on P0 F1 ($q^* = 10\%$). Worst-source recall (left) and NDCG (right) are plotted against budget factor b for the best single-source baselines and three source-robust aggregation strategies. The vertical dashed line marks the equal-budget analysis at $b = 1.0$. Aggregation changes the coverage-quality trade-off but does not validate physical truth. See Note 1 for the distinction between material-level recall and group-mean paired differences.